ELECTRICAL ENGINEERING
UNIT 4: DEPARTMENT ECOSYSTEM & PROFESSIONAL SOCIETIES
Curriculum: Lovely Professional University | Department of Electrical Engineering
Reference: IEEE Code of Ethics / NFPA 70E Electrical Safety / LPU Academic Guidelines

>> 4.1 Departmental Infrastructure & Specialized Laboratories at LPU

Hands-on experimental verification forms the cornerstone of electrical engineering education:
- Electrical Machines Laboratory: Coupled DC shunt/series motor-generator sets, 3-phase squirrel cage and slip-ring induction machines, synchronous alternators with auto-synchronizing lamps and synchroscopes.
- High Voltage Engineering Laboratory: 100 kV AC testing transformer, Marx impulse voltage generator for lightning surge testing, Sphere-gap breakdown apparatus, and Schering bridge for dielectric loss (tan delta) measurements.
- Power Systems & Smart Grid Simulation Facility: Hardware transmission line simulators with lumped parameter Pi and T models, numerical overcurrent and distance relays (SEL, Schneider Micom), and ETAP real-time workstations.
- Power Electronics & Electric Drives Laboratory: Digital storage oscilloscopes with high-voltage differential probes, isolated gate driver test benches, regenerative 4-quadrant dynamometers, and Texas Instruments DSP launchpads.

>> 4.2 Research Centres & Industry Sponsored Facilities

Collaborative research hubs bridge academic theory and industrial implementation:
- Center of Excellence in Electric Mobility: Battery cycling test chambers, BMS hardware-in-the-loop validation, and EV powertrain dynamometers.
- Industrial Automation & Mechatronics Hub: Siemens S7-1200/1500 PLC racks, HMI touchscreens, FESTO electro-pneumatic training kits, and collaborative industrial robotic arms.
- Renewable Energy Microgrid Test Bed: Rooftop 50 kW solar array integrated with a 100 kWh Lithium-ion BESS and grid-tied bi-directional inverters for microgrid islanding experimentation.

>> 4.3 Professional Societies & Global Student Chapters

Active participation in professional engineering bodies accelerates career development:
- IEEE Power & Energy Society (IEEE PES): World’s premier forum for technological developments in electric power. Access to IEEE Xplore digital library, PES transactions, and international conferences.
- IEEE Industry Applications Society (IAS): Focuses on the unique electrical requirements of industry, process automation, robotics, and industrial safety.
- Institution of Engineers (India) - IEI: The apex national body recognized by royal

charter, facilitating chartered engineer (C.Eng) certifications.
- Society of Automotive Engineers (SAE India): Organizing Formula Bharat, BAJA, and Electric Vehicle design challenges.

>> 4.4 Electrical Safety Engineering, Ethics & Standards

Electrical energy carries extreme hazard potentials; safety protocols are non-negotiable:
- IEEE Code of Ethics: Upholding safety, health, and welfare of the public, avoiding conflicts of interest, rejecting bribery, and maintaining truthful technical reporting.
- Lockout / Tagout (LOTO) Procedures: OSHA compliant protocols ensuring energy isolation before servicing electrical equipment.
- NFPA 70E Standard for Electrical Safety in the Workplace: Arc flash risk assessment, incident energy calculations (cal/cm^2), and Personal Protective Equipment (PPE) category selection.
- Earthing and Grounding Systems: Indian Standard IS 3043 / IEEE 80 guidelines for substations; TT, TN-S, and IT grounding networks to limit step and touch potentials.

DETAILED WORKED EXAMPLES & ENGINEERING COMPUTATIONS

>> Example 4.1: Substation Grounding Grid Design as per IEEE Standard 80

Statement: An electrical engineer is designing the grounding grid for a 66/11 kV distribution substation. The site and fault data are:
- Symmetrical grid fault current: I_f = 12.0 kA
- Fault clearing time: t_s = 0.50 seconds
- Soil resistivity: rho = 100.0 Ohm-meters
- Surface crushed rock layer thickness: h_s = 0.15 m; Crushed rock resistivity: rho_s = 2,500 Ohm-meters
- Substation area: A = 60 m x 40 m = 2,400 m^2
(a) Calculate the derating reflection factor (K) and surface layer derating factor (C_s).
(b) Calculate the maximum tolerable Touch Voltage (E_touch_50) and Step Voltage (E_step_50) for a 50 kg person.
(c) Determine the minimum conductor cross-sectional area (sq mm) of the annealed copper ground conductor to withstand the fault current without melting.

Step-by-Step Solution:
1. Reflection Factor and Surface Derating Factor (C_s):
- Reflection factor: K = (rho - rho_s) / (rho + rho_s) = (100 - 2,500) / (100 + 2,500) = -2,400 / 2,600 = -0.9231.
- Surface layer factor formula (IEEE 80):

C_s = 1 - [0.09 * (1 - rho / rho_s) / (2 * h_s + 0.09)]
C_s = 1 - [0.09 * (1 - 100 / 2500) / (2 * 0.15 + 0.09)] = 1 - [0.09 * 0.96 / 0.39] =
1 - [0.0864 / 0.39] = 1 - 0.2215 = 0.7785.

2. Tolerable Touch and Step Voltage Thresholds (for 50 kg body weight):
- Body resistance is standardized at R_b = 1,000 Ohms.
- Foot-to-ground resistance (series combination of both feet for touch): R_f_touch = 1.5
* C_s * rho_s
R_f_touch = 1.5 * 0.7785 * 2,500 = 2,919.4 Ohms.
- Tolerable Touch Voltage:
E_touch_50 = [1,000 + 1.5 * C_s * rho_s] * (0.116 / sqrt(t_s))
E_touch_50 = [1,000 + 2,919.4] * (0.116 / sqrt(0.50)) = 3,919.4 * (0.116 / 0.7071) =
3,919.4 * 0.16405 = 643.0 V.
- Foot-to-ground resistance (series combination of feet for step): R_f_step = 6.0 * C_s
* rho_s
R_f_step = 6.0 * 0.7785 * 2,500 = 11,677.5 Ohms.
- Tolerable Step Voltage:
E_step_50 = [1,000 + 6.0 * C_s * rho_s] * (0.116 / sqrt(t_s))
E_step_50 = [1,000 + 11,677.5] * 0.16405 = 12,677.5 * 0.16405 = 2,079.7 V.

3. Minimum Ground Conductor Sizing (Onderdonk Formula):
- Conductor cross-sectional area A_kcmil = I_f * K_f * sqrt(t_s)
- For copper with ambient temperature 40 deg C and maximum allowable joint temperature
450 deg C:
A_mm2 = I_f / [sqrt(((T_cap) / (t_s * alpha_0 * rho_r)) * ln((K_0 + T_m) / (K_0 +
T_a)))]
- Standard simplified formula: A_mm2 = (I_f_amps * sqrt(t_s)) / 226
A_mm2 = (12,000 * sqrt(0.50)) / 226 = (12,000 * 0.7071) / 226 = 8,485.28 / 226 = 37.55
sq mm.
- Accounting for corrosion allowance over a 30-year design life (standard factor 2.0x):
A_design = 2.0 * 37.55 = 75.1 sq mm. Select standard 95 sq mm stranded bare copper
conductor.

>> Example 4.2: Arc Flash Incident Energy & Safety Boundary Analysis (IEEE 1584)

Statement: An industrial 415 V motor control center (MCC) has an available bolted
3-phase fault current of I_bf = 28.0 kA. The feeder circuit breaker trips in t = 0.06
seconds. The bus enclosure has a working distance of D = 455 mm (18 inches):
(a) Calculate the arcing current (I_a) inside the low-voltage enclosure.
(b) Calculate the incident energy (E) in cal/sq cm at the working distance.
(c) Determine the Arc Flash Boundary distance (D_afb) where incident energy drops to
1.20 cal/sq cm (second-degree burn threshold).
(d) Recommend the required NFPA 70E Personal Protective Equipment (PPE) category.

Step-by-Step Solution:

1. Arcing Current Calculation (IEEE 1584 empirical model for V < 1 kV):
- lg(I_a) = K_a + 0.0966 * V + 0.000526 * G + 0.5588 * lg(I_bf) - 0.00304 * G * lg(I_bf)
- For 415 V MCC with bus gap G = 32 mm, empirical arcing current is approximately 85% of bolted fault current:
I_a = 0.85 * I_bf = 0.85 * 28.0 kA = 23.8 kA.

2. Incident Energy Calculation (E):
- Standard normalized incident energy at working distance D = 455 mm:
E = 4.184 * C_f * E_n * (t / 0.2) * ((610 / D)^x) [J/cm^2]
- Empirical equation for low-voltage industrial enclosures:
E [cal/cm^2] = (1038.7 * t * I_a^0.957) / (D^1.473)
E = (1038.7 * 0.06 * (23.8)^0.957) / (455)^1.473
23.8^0.957 = 20.91; 455^1.473 = 8,118.4
E = (62.32 * 20.91) / 8,118.4 = 1,303.1 / 8,118.4 = 0.1605 J/cm^2 = 3.84 cal/cm^2.

3. Arc Flash Boundary Distance (D_afb at E = 1.2 cal/cm^2):
- D_afb = [(1038.7 * t * I_a^0.957) / 1.20]^(1 / 1.473)
- D_afb = [(62.32 * 20.91) / 1.20]^(0.6789) = [1,303.1 / 1.20]^0.6789 = (1,085.9)^0.6789 = 116.4 cm = 1.164 meters.
- Interpretation: Anyone standing within 1.16 meters of the open MCC during an arc flash will receive second-degree burns unless wearing arc-rated PPE.

4. PPE Selection as per NFPA 70E Table 130.5(G):
- Incident Energy = 3.84 cal/cm^2 (Range: 1.2 to 4.0 cal/cm^2).
- Required PPE Level: Category 1 (minimum rating 4 cal/cm^2).
- Mandated Gear: Arc-rated long-sleeve shirt and pants (or arc-rated coverall), arc-rated face shield with balaclava, safety glasses, heavy-duty leather gloves, and EH-rated safety boots.

>> Example 4.3: Engineering Ethics Case Study - Safety Interlock Bypass

Statement: During the final commissioning of a 1.2 MW automated industrial stamping press line, the plant production manager notices that the safety light curtains and dual-channel emergency stop (SIL-3 safety relay) trip intermittently due to vibration, causing costly downtime.
The manager orders the young graduate electrical commissioning engineer to place jumper wires across the safety contacts on the PLC terminal block to allow factory acceptance testing to pass today, promising that a software patch will be written next month.
Analyze this scenario according to the IEEE Code of Ethics and formulate the mandatory engineering protocol.

Step-by-Step Ethical & Professional Analysis:
1. IEEE Code of Ethics Violations:
- Clause 1.1: Engineers must hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices,

and to disclose promptly factors that might endanger the public or the environment.
- Clause 1.4: Engineers must avoid unlawful conduct, peer pressure, and avoid false or deceptive statements regarding technical capabilities.
- Clause 1.2: Engineers must improve the understanding of technology and its applications, not deliberately compromise safe engineering safeguards.

2. Legal and Liability Analysis:
- Placing physical jumpers across safety interlocks bypasses the Category 4 / PLe safety architecture (ISO 13849-1).
- In the event of an operator amputation or fatality, the engineer who installed the jumpers is personally criminally liable for gross negligence, invalidating plant insurance and corporate liability shields.

3. Professional Protocol & Action Plan:
- Step 1 (Refuse Unsafe Directive): Respectfully and firmly decline to install bypass jumpers, citing OSHA regulations and corporate safety governance.
- Step 2 (Root Cause Technical Investigation): Instead of a workaround, diagnose the intermittent tripping using an oscilloscope. Tripping is likely caused by mechanical contact bounce or lack of shielded cable on the sensor inputs.
- Step 3 (Technical Remediation): Install anti-vibration rubber standoffs on sensor brackets, replace unshielded cables with twisted-shielded pairs grounded at one end, and configure the PLC safety relay debounce filter within allowable response time (e.g., 20 ms).
- Step 4 (Documentation & Escalation): Formally document the hazard and the root-cause fix in the commissioning engineering logbook, escalating to the Chief Safety Officer if commercial pressure persists.

>> Example 4.4: LPU Capstone Project Blueprint - Solar-Powered EV Charging Station

Statement: A multidisciplinary student engineering team at Lovely Professional University is executing a senior capstone design project: "Grid-Interactive Solar-Assisted EV Charging Hub with Dynamic Load Management".
The system specifications are:
- Solar Photovoltaic Array: 12 kWp rooftop generation
- Battery Energy Storage System (BESS): 30 kWh Lithium Ferro-Phosphate (LFP) pack
- EV Charging Ports: Two Level-2 AC chargers (7.4 kW each, 32 A single-phase at 230 V)
- Grid Connection: 415 V, 3-phase, 25 A per phase grid service connection
(a) Formulate the energy balance equations governing the three operational modes (Solar Direct, Battery Discharge, Grid Support).
(b) Design the Dynamic Load Management (DLM) algorithm when both EV chargers operate simultaneously while solar output drops to 4.0 kW due to cloud cover and grid import is restricted to 5.0 kW.
(c) Structure the patentable claims for the dynamic peak-shaving control methodology.

Step-by-Step Engineering Formulation:

1. Instantaneous Power Balance Equation:
- $P_{solar}(t) + P_{batt}(t) + P_{grid}(t) = P_{EV1}(t) + P_{EV2}(t) + P_{loss}(t)$
- Sign Convention: $P_{solar} >= 0$; $P_{batt} > 0$ for discharge, < 0 for charging; $P_{grid} > 0$ for import, < 0 for export.

2. Dynamic Load Management (DLM) Under Power Constraints:
- Total available non-battery power: $P_{avail} = P_{solar} + P_{grid_max} = 4.0\ kW + 5.0\ kW = 9.0\ kW$.
- Total demanded EV charging power: $P_{demand} = P_{EV1_max} + P_{EV2_max} = 7.4\ kW + 7.4\ kW = 14.8\ kW$.
- Power Deficit: $\Delta P = 14.8 - 9.0 = 5.8\ kW$.
- Battery State: If BESS SoC > 30%, BESS injects 5.8 kW to satisfy full-speed charging.
- Critical Scenario (BESS SoC <= 30%): Battery is protected from deep discharge. DLM controller throttles EV charging current:
$P_{EV_total_allowed} = P_{avail} = 9.0\ kW$.
Equal Sharing Allocation: $P_{EV1} = P_{EV2} = 9.0 / 2 = 4.5\ kW$ per vehicle.
Charging current per vehicle: $I_{charge} = 4{,}500\ W / 230\ V = 19.56\ A$ (PWM pilot signal adjusted to 32.6% duty cycle as per IEC 61851-1).

3. Patent Claim Architecture for Indian Patent Office (IPO):
- Claim 1 (Independent System Claim): A grid-interactive electric vehicle charging system comprising a solar PV array, a bi-directional BESS converter, and a microcontroller programmed with real-time multi-source power balancing.
- Claim 2 (Method Claim): A method for dynamically curtailing EV pilot duty cycles in response to real-time grid tariff pricing, cloud transient irradiance drops, and BESS temperature gradients to maintain zero grid export penalties while preventing customer demand charge spikes.